

and

$$u = 0 = a \ln \left(\frac{1}{k_j} \right) + C$$

$$C = -a \ln \left(\frac{1}{k_j} \right)$$

Substituting for C in the equation for u , we obtain

$$\begin{aligned} u &= a \ln \left(\frac{1}{k} \right) - a \ln \left(\frac{1}{k_j} \right) \\ &= a \ln \left(\frac{k_j}{k} \right) \end{aligned}$$

which is the Greenberg model given in Eq. 6.19. Similarly, if m is allowed to be 0 and $\ell = 2$, we obtain the Greenshields model.

6.3 SHOCK WAVES IN TRAFFIC STREAMS

The fundamental diagram of traffic flow for two adjacent sections of a highway with different capacities (maximum flows) is shown in Figure 6.8. This figure describes the phenomenon of backups and queuing on a highway due to a sudden reduction of the capacity of the highway (known as a *bottleneck condition*). The sudden reduction in capacity could be due to a crash, reduction in the number of lanes, restricted bridge sizes, work zones, a signal turning red, and so forth, creating a situation where the

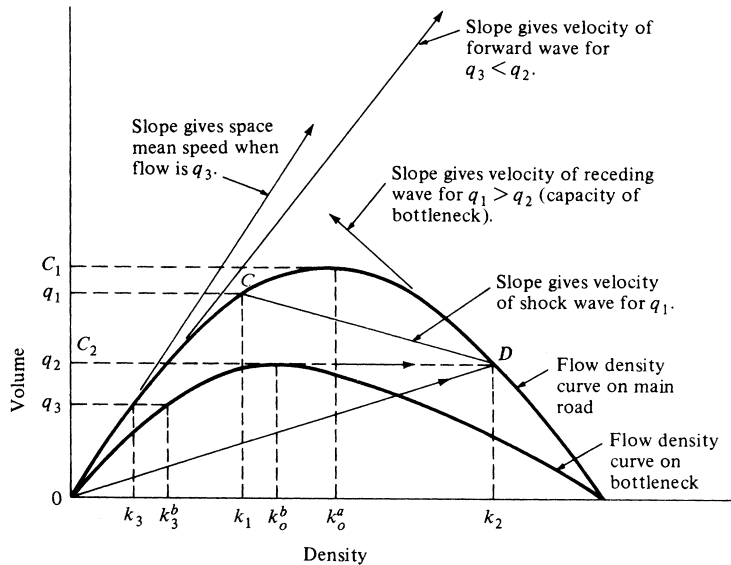


Figure 6.8 Kinematic and Shock Wave Measurements Related to Flow-Density Curve

capacity on the highway suddenly changes from C_1 to a lower value of C_2 , with a corresponding change in optimum density from k_o^a to a value of k_o^b .

When such a condition exists and the normal flow and density on the highway are relatively large, the speeds of the vehicles will have to be reduced while passing the bottleneck. The point at which the speed reduction takes place can be approximately noted by the turning on of the brake lights of the vehicles. An observer will see that this point moves upstream as traffic continues to approach the vicinity of the bottleneck indicating an upstream movement of the point at which flow and density change. This phenomenon is usually referred to as a *shock wave* in the traffic stream. The phenomenon also exists when the capacity suddenly increases but in this case, the speeds of the vehicles tend to increase as the vehicles pass the section of the road where the capacity increases.

6.3.1 Types of Shock Waves

Several types of shock waves can be formed, depending on the traffic conditions that lead to their formation. These include *frontal stationary*, *backward forming*, *backward recovery*, *rear stationary* and *forward recovery* shock waves.

Frontal stationary shock waves are formed when the capacity suddenly reduces to zero at an approach or set of lanes having the red indication at a signalized intersection or when a highway is completely closed because of a serious incident. In this case, a frontal stationary shock wave is formed at the stop line of the approach or lanes that have a red signal indication. This type occurs at the location where the capacity is reduced to zero. For example, at a signalized intersection, the red signal indicates that traffic on the approach or set of lanes cannot move across the intersection, which implies that the capacity is temporarily reduced to zero resulting in the formation of a frontal stationary shock wave as shown in Figure 6.9 on the next page.

Backward forming shock waves are formed when the capacity is reduced below the demand flow rate resulting in the formation of a queue upstream of the bottleneck. The shock wave moves upstream with its location at any time indicating the end of the queue at that time. This may occur at the approach of a signalized intersection when the signal indication is red, as shown in Figure 6.9, or at a location of a highway where the number of lanes is reduced.

Backward recovery shock waves are formed when the demand flow rate becomes less than the capacity of the bottleneck or the restriction causing the capacity reduction at the bottleneck is removed. For example, when the signals at an approach or set of lanes on a signalized intersection change from red to green, the traffic flow restriction is removed, and traffic on that approach or set of lanes is free to move across the intersection, causing a backward recovery shock wave as shown in Figure 6.9. The intersection of the backward forming shock wave and the backward recovery shock wave indicates the end of the queue shown as point *T* in Figure 6.9.

Rear stationary and forward recovery shock waves are formed when demand flow rate upstream of a bottleneck is first higher than the capacity of the bottleneck and then the demand flow rate reduces to the capacity of the bottleneck. For example, consider a four-lane (one direction) highway that leads to a two-lane tunnel in an urban area as shown in Figure 6.10. During the off-peak period when the demand

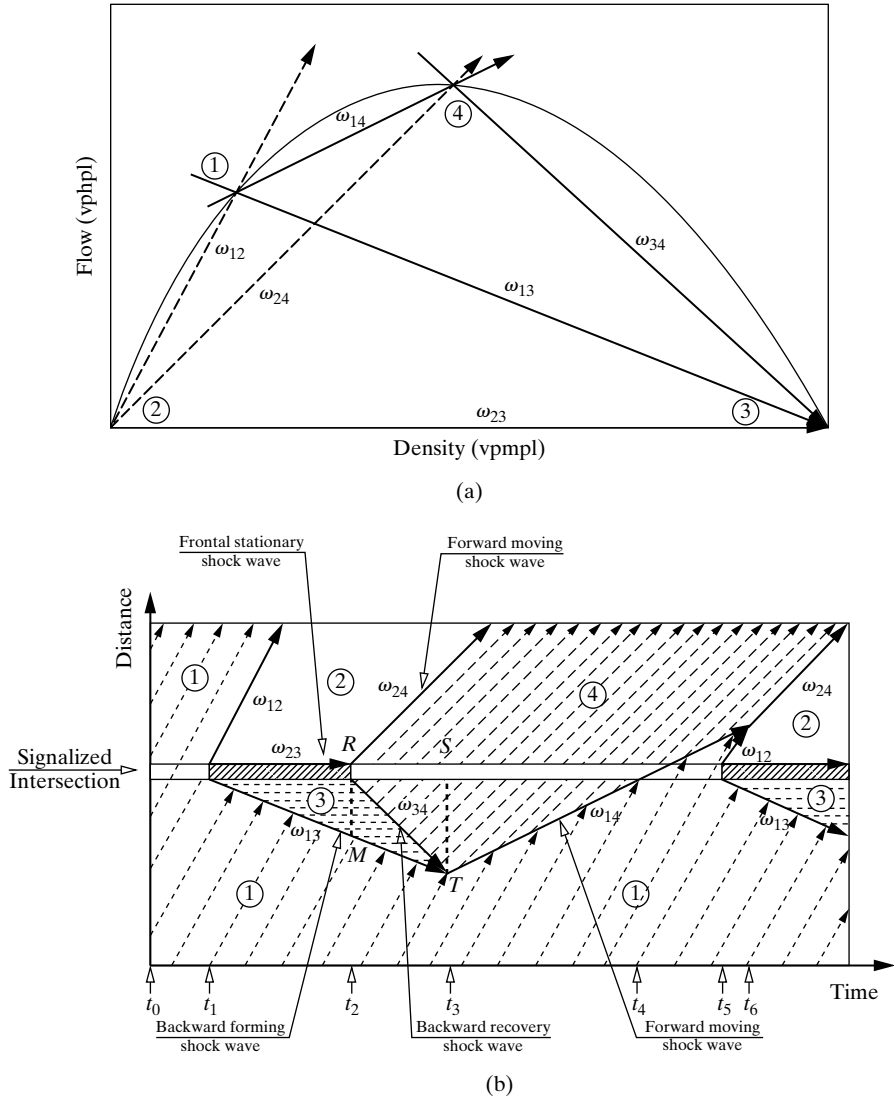


Figure 6.9 Shock Wave at Signalized Intersection

capacity is less than the tunnel capacity, no shock wave is formed. However, when the demand capacity becomes higher than the tunnel capacity during the peak hour, a backward forming shock wave is formed. This shock wave continues to move upstream of the bottleneck as long as the demand flow is higher than the tunnel capacity as shown in Figure 6.10. However, as the end of the peak period approaches, the demand flow rate tends to decrease until it is the same as the tunnel capacity. At this point, a rear stationary shock wave is formed until the demand flow becomes less than the tunnel capacity resulting in the formation of a forward recovery shock wave as shown in Figure 6.10.

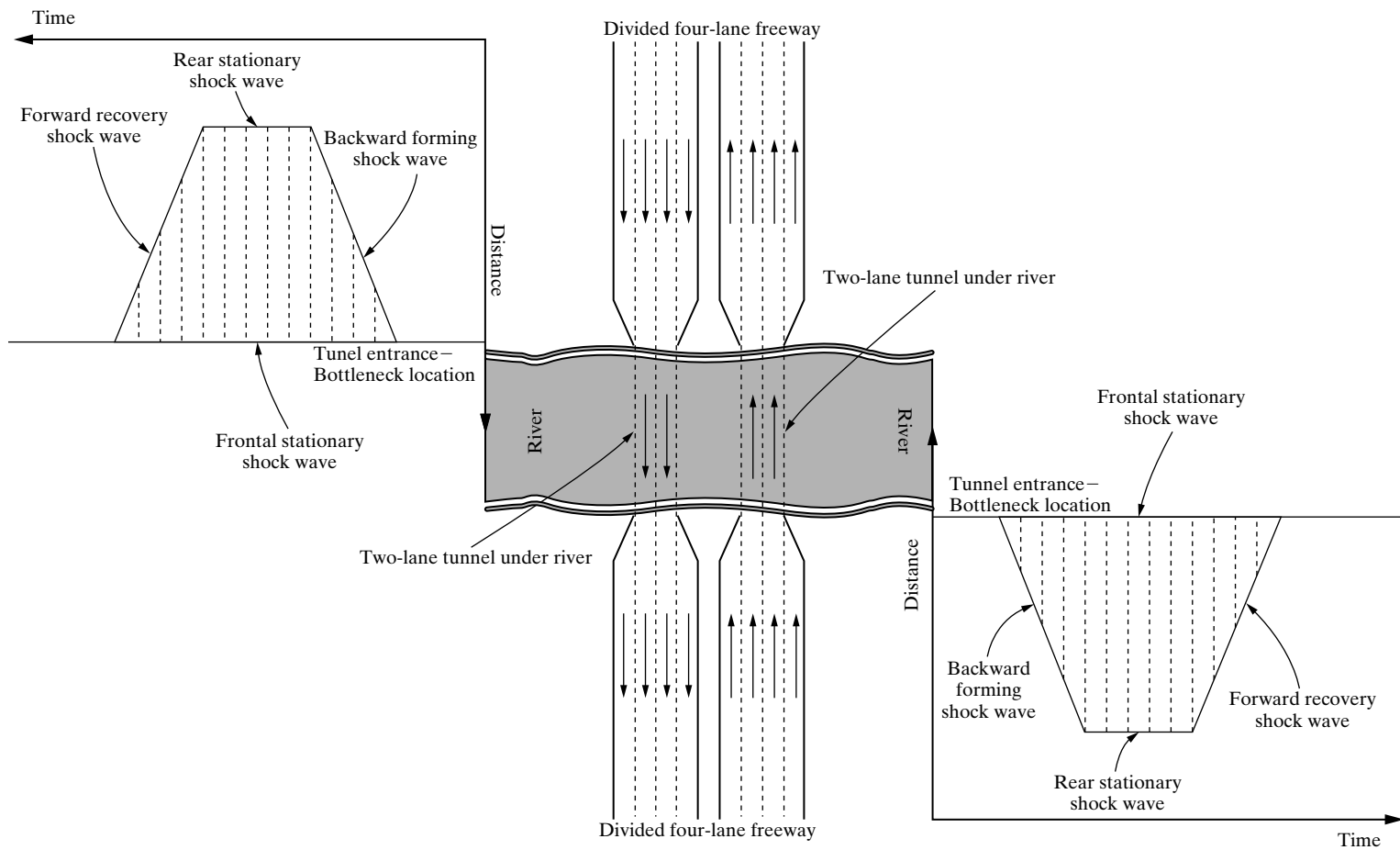


Figure 6.10 Shock Waves Due to a Bottleneck

6.3.2 Velocity of Shock Waves

Let us consider two different densities of traffic, k_1 and k_2 , along a straight highway as shown in Figure 6.11 where $k_1 > k_2$. Let us also assume that these densities are separated by the line w representing the shock wave moving at a speed u_w . If the line w moves in the direction of the arrow (that is, in the direction of the traffic flow), u_w is positive.

With u_1 equal to the space mean speed of vehicles in the area with density k_1 (section P), the speed of the vehicle in this area relative to line w is

$$u_{r_1} = (u_1 - u_w)$$

The number of vehicles crossing line w from area P during a time period t is

$$N_1 = u_{r_1} k_1 t$$

Similarly, the speed of vehicles in the area with density k_2 (section Q) relative to line w is

$$u_{r_2} = (u_2 - u_w)$$

and the number of vehicles crossing line w during a time period t is

$$N_2 = u_{r_2} k_2 t$$

Since the net change is zero—that is, $N_1 = N_2$ and $(u_1 - u_w)k_1 = (u_2 - u_w)k_2$ —we have

$$u_2 k_2 - u_1 k_1 = u_w (k_2 - k_1) \quad (6.32)$$

If the flow rates in sections P and Q are q_1 and q_2 , respectively, then

$$q_1 = k_1 u_1 \quad q_2 = k_2 u_2$$

Substituting q_1 and q_2 for $k_1 u_1$ and $k_2 u_2$ in Eq. 6.32 gives

$$q_2 - q_1 = u_w (k_2 - k_1)$$

That is

$$u_w = \frac{q_2 - q_1}{k_2 - k_1} \quad (6.33)$$

which is also the slope of the line CD shown in Figure 6.8. This indicates that the velocity of the shock wave created by a sudden change of density from k_1 to k_2 on a traffic stream is the slope of the chord joining the points associated with k_1 and k_2 on the volume density curve for that traffic stream.

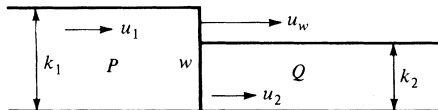


Figure 6.11 Movement of Shock Wave Due to Change in Densities

6.3.3 Shock Waves and Queue Lengths Due to a Red Phase at a Signalized Intersection

Figure 6.9 also shows the traffic conditions that exist at an approach of a signalized intersection when the signal indication is green then changes to red at the end of the green phase (start of the red phase) and changes to green again at the end of the red phase (start of the green phase). When the signal indication is green, the flow is normal as shown in section 1. When the signals change to red at time t_1 , two new conditions are formed immediately.

Flow from this approach is stopped creating section 2, immediately downstream of the stop line with a density of zero and flow of zero. At the same time, all vehicles immediately upstream of the stop line are stationary, forming section 3, where the flow is zero and the density is the jam density. This results in the formation of the frontal stationary shock wave with velocity ω_{23} and the backward forming shock wave with velocity ω_{13} .

At the end of the red phase at time t_2 when the signal indication changes to green again, the flow rate at the stop line changes from zero to the saturation flow rate (see Chapter 8 for definition) as shown in section 4. This results in the forward moving shock wave ω_{24} . The queue length at this time—that is at the end of the red phase—is represented by the line RM . Also at this time, the backward recovery shock wave with velocity of ω_{34} is formed that releases the queue as it moves upstream of the stop line. The intersection of the backward forming and backward recovery shock waves at point T and time t_3 indicates the position where the queue is completely dissipated with the maximum queue length being represented by the line ST . The backward forming and backward recovery shock waves also terminate at time t_3 and a new forward moving shock wave with velocity ω_{14} is formed.

When the forward moving shock wave crosses the stop line, at time t_4 , the flow changes at the stop line from the saturated flow rate to the original flow rate in section 1 and this continues until time t_5 when the signals change again to red.

Using Eq. 6.33, we can determine expressions for the velocities of the different shock waves and the queue lengths:

$$\text{The shock wave velocity } \omega_{12} = \frac{q_2 - q_1}{k_2 - k_1} = \frac{q_1 - q_2}{k_1 - k_2} = \frac{q_1 - 0}{k_1 - 0} = u_1 \quad (6.34)$$

$$\text{The shock wave velocity } \omega_{13} = \frac{q_1 - q_3}{k_1 - k_3} = \frac{q_1 - 0}{k_1 - k_j} = \frac{q_1}{k_1 - k_j} \quad (6.35)$$

$$\text{The shock wave velocity } \omega_{23} = \frac{q_2 - q_3}{k_2 - k_3} = \frac{0 - 0}{0 - k_j} = \frac{0}{k_j} = 0 \quad (6.36)$$

This confirms that this wave is a stationary wave.

$$\text{The shock wave velocity } \omega_{24} = \frac{q_2 - q_4}{k_2 - k_4} = \frac{0 - q_4}{0 - k_4} = u_4 \quad (6.37)$$

$$\text{The shock wave velocity } \omega_{34} = \frac{q_3 - q_4}{k_3 - k_4} = \frac{0 - q_4}{k_j - k_4} = \frac{-q_4}{k_j - k_4} \quad (6.38)$$

$$\begin{aligned} \text{The length of the queue at the end of the red signal} &= r \times \omega_{13} \\ &= \frac{rq_1}{k_1 - k_j} \end{aligned} \quad (6.39)$$

where r = the length of the red signal indication.

The maximum queue length $\overline{ST}$ can be determined from Figure 6.9 from where it can be seen that $\omega_{34} = \tan \gamma = \frac{\overline{ST}}{\overline{RS}}$ which gives $\overline{RS} = \frac{\overline{ST}}{\tan \gamma}$. Also, ω_{13} is $\tan \varphi$:

$$\begin{aligned} \tan \varphi &= \frac{\overline{ST}}{r + \overline{RS}} \\ \overline{ST} &= \tan \varphi (r + \overline{RS}) \\ \overline{ST} &= \tan \varphi \left(r + \frac{\overline{ST}}{\tan \gamma} \right) \\ r &= \frac{\overline{ST}}{\tan \varphi} - \frac{\overline{ST}}{\tan \gamma} \\ \overline{ST} &= \frac{r}{\frac{1}{\tan \varphi} - \frac{1}{\tan \gamma}} \\ \overline{ST} &= \frac{r \tan \varphi \tan \gamma}{\tan \gamma - \tan \varphi} \\ \overline{ST} &= \frac{r \omega_{13} \omega_{34}}{\omega_{34} - \omega_{13}} \end{aligned} \quad (6.40)$$

The additional time $\overline{RS}$ (i.e., $t_3 - t_2$) after the end of the red signal it takes for the maximum queue to be formed can be obtained from the expression $\tan \varphi = \frac{\overline{ST}}{r + \overline{RS}}$, which gives

$$\begin{aligned} \overline{RS} &= \frac{\overline{ST}}{\tan \varphi} - r \\ &= \frac{r \omega_{13} \omega_{34}}{\omega_{13}(\omega_{34} - \omega_{13})} - r \\ &= \frac{r \omega_{13}}{\omega_{13} - \omega_{34}} \end{aligned} \quad (6.41)$$

Example 6.4 Queue Lengths at a Signalized Intersection

The southbound approach of a signalized intersection carries a flow of 1000 veh/h/ln at a velocity of 50 mi/h. The duration of the red signal indication for this approach is 15 sec. If the saturation flow is 2000 veh/h/ln with a density of 75 veh/ln, the jam density is 150 veh/mi, determine the following:

- The length of the queue at the end of the red phase
- The maximum queue length
- The time it takes for the queue to dissipate after the end of the red indication.

Solution:

- Determine speed of backward forming shock wave ω_{13} when signals turn to red. Use Eq. 6.33.

$$\begin{aligned}\omega_w &= \frac{q_2 - q_1}{k_2 - k_1} \\ \omega_{13} &= \frac{q_1 - q_3}{k_1 - k_3} \\ q_1 &= 1000 \text{ veh/h/ln} \\ q_3 &= 0 \text{ veh/h/ln} \\ k_1 &= \frac{1000}{50} = 20 \text{ veh/mi (see Eq. 6.7.)} \\ \omega_{13} &= \frac{1000 - 0}{20 - 150} \text{ mi/h} = -7.69 \text{ mi/h} \\ &= -7.69 \times 1.47 \text{ ft/sec} = -11.31 \text{ ft/sec}\end{aligned}$$

Length of queue at end of red phase = $15 \times 11.31 = 169.65 \text{ ft}$

- Determine speed of backward recovery wave velocity. Use Eq. 6.33.

$$\omega_{34} = \frac{q_3 - q_4}{k_3 - k_4} = \frac{0 - 2000}{150 - 75} = -26.67 \text{ mi/h} = -26.67 \times 1.47 = -39.2 \text{ ft/sec}$$

- Determine the maximum queue length. Use Eq. 6.40.

$$\text{Maximum queue length} = \frac{r\omega_{13}\omega_{34}}{\omega_{34} - \omega_{13}} = \frac{15 \times 11.31 \times 39.2}{39.2 - 11.31} = 238.45 \text{ ft}$$

6.3.4 Shock Waves and Queue Lengths Due to Temporary Speed Reduction at a Section of Highway

Let us now consider the situation where the normal speed on a highway is temporarily reduced at a section of a highway where the flow is relatively high but lower than its capacity. For example, consider a truck that enters a two-lane highway at time t_1 and

traveling at a much lower speed than the speed of the vehicles driving behind it. The truck travels for some time on the highway and eventually leaves the highway at time t_2 . If the traffic condition is such that the vehicles cannot pass the truck, the shock waves that will be formed are shown in Figure 6.12. The traffic conditions prior to the truck entering the highway at time t_1 is depicted as section 1.

At time t_1 , vehicles immediately behind the truck will reduce their speed to that of the truck. This results in an increased density immediately behind the truck resulting in traffic condition 2. The moving shock wave with a velocity of ω_{12} is formed. Also, because vehicles ahead of the truck will continue to travel at their original

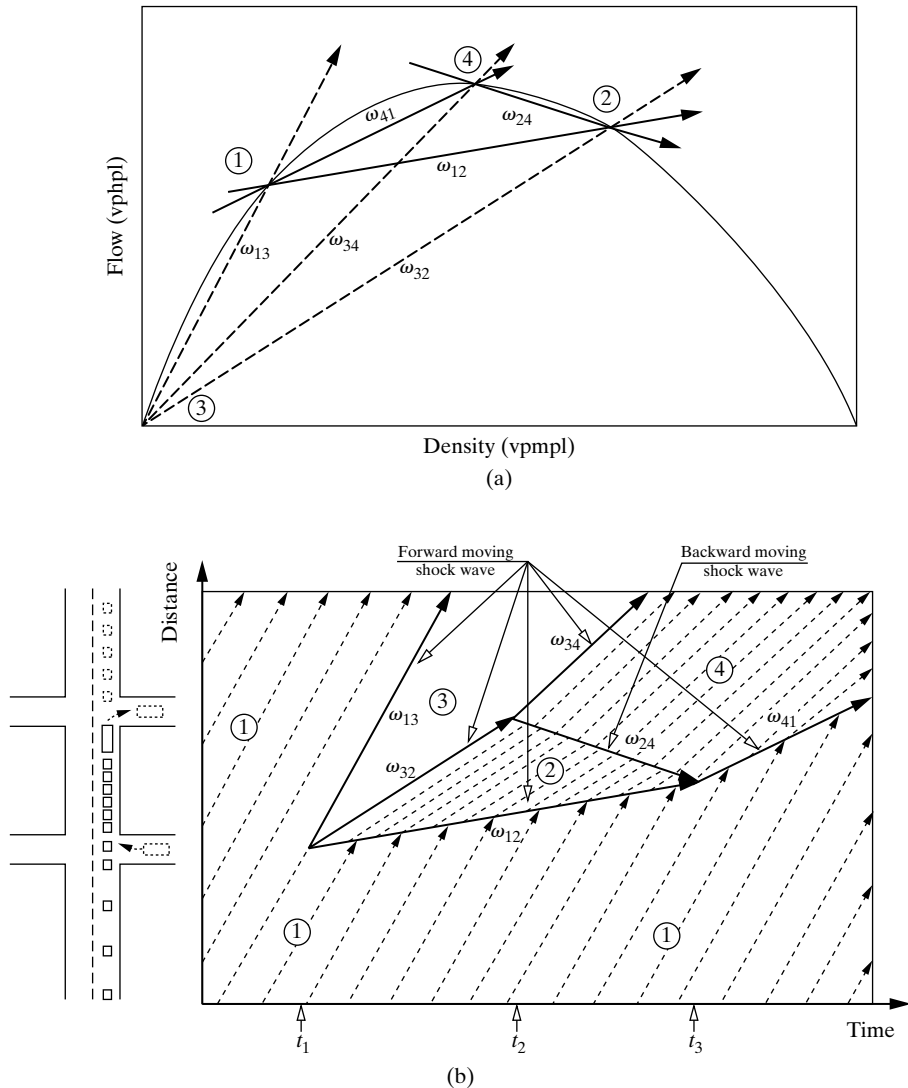


Figure 6.12 Shock Wave Created By Slow Traffic

speed, a section on the highway just downstream of the truck will have no vehicles thereby creating traffic condition 3. This also results in the formation of the forward moving shock waves with velocities of ω_{13} , and ω_{32} . At time t_2 when the truck leaves the highway, the flow will be increased to the capacity of the highway with traffic condition 4. This results in the formation of a backward moving shock wave velocity ω_{24} and a forward moving shock wave with velocity ω_{34} . At time t_3 , shock waves with velocities ω_{12} and ω_{24} coincide resulting in a new forward moving shock wave with a velocity ω_{41} . It should be noted that the actual traffic conditions 2 and 4 depend on the original traffic condition 1 and the speed of the truck.

Example 6.5 Length of Queue Due to a Speed Reduction

The volume at a section of a two-lane highway is 1500 veh/h in each direction and the density is about 25 veh/mi. A large dump truck loaded with soil from an adjacent construction site joins the traffic stream and travels at a speed of 10 mi/h for a length of 2.5 mi along the upgrade before turning off onto a dump site. Due to the relatively high flow in the opposite direction, it is impossible for any car to pass the truck. Vehicles just behind the truck therefore have to travel at the speed of the truck which results in the formation of a platoon having a density of 100 veh/mi and a flow of 1000 veh/h. Determine how many vehicles will be in the platoon by the time the truck leaves the highway.

Solution: Use Eq. 6.33 to obtain the wave velocity.

$$\begin{aligned} u_w &= \frac{q_2 - q_1}{k_2 - k_1} \\ u_w &= \frac{1000 - 1500}{100 - 25} \\ &= -6.7 \text{ mi/h} \end{aligned}$$

Knowing that the truck is traveling at 10 mi/h, the speed of the vehicles in the platoon is also 10 mi/h and that the shock wave is moving backward relative to the road at 6.7 mi/h, determine the growth rate of the platoon.

$$10 - (-6.7) = 16.7 \text{ mi/h}$$

Calculate the time spent by the truck on the highway— $2.5/10 = 0.25$ h—to determine the length of the platoon by the time the truck leaves the highway.

$$0.25 \times 16.7 = 4.2 \text{ mi}$$

Use the density of 100 veh/mi to calculate the number of vehicles in the platoon.

$$100 \times 4.2 = 420 \text{ vehicles}$$